

Advanced Empirical Finance - Part 2

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Table of contents

1	Exercise 1: Choice of the Realized Variance Estimator	1
1.1	Load Packages and Data	1
1.2	Prepare AAPL Data	1
1.3	Function for Daily Realized Variance	1
1.4	Compute Realized Variance for All Sampling Frequencies	2
1.5	Summary Statistics	2
1.6	Volatility Signature Plot	2
1.7	Interpretation	3

1 Exercise 1: Choice of the Realized Variance Estimator

1.1 Load Packages and Data

1.2 Prepare AAPL Data

1.3 Function for Daily Realized Variance

Following the notation from the lecture slides, let X_t denote the log-price process. Intraday log-returns are computed as

$$r_t = X_t - X_{t-\Delta},$$

where Δ denotes the sampling interval in minutes. For each trading day and each sampling frequency $\Delta \in 1, 2, 5, 10, 15, 30, 60$, realized variance is computed as the sum of squared intraday returns:

$$RV^{(n)} = \sum_{i=0}^{n-1} r_{i,n}^2.$$

This corresponds to the realized variance estimator introduced in the lecture notes. Under a continuous diffusion process without jumps, realized variance converges to the integrated variance as the sampling frequency increases.

The dataset contains minute-level transaction prices. Because a stock may not trade in every minute, missing observations are handled using previous-tick interpolation (forward filling). Missing prices are replaced by the most recently observed transaction price, implying a zero return during periods without new trades.

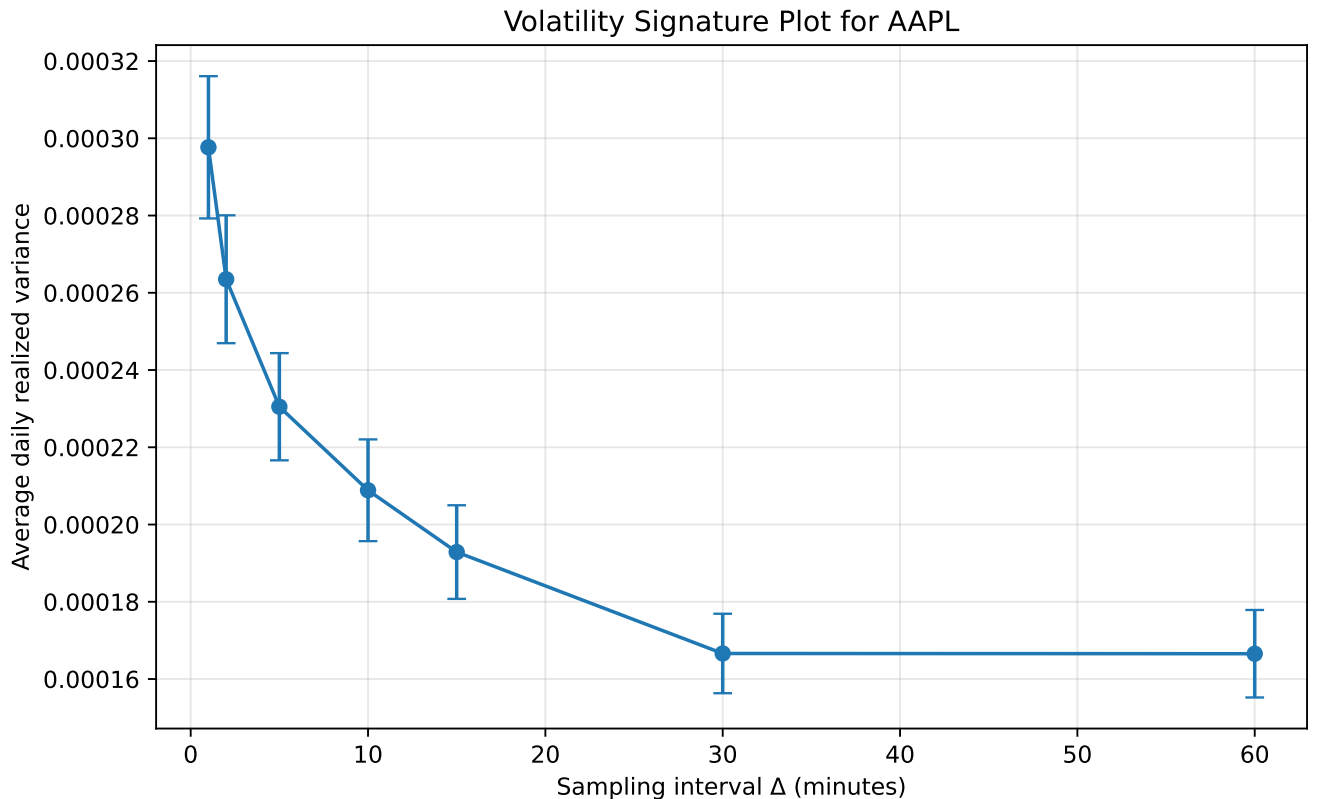
For each sampling interval, daily realized variance is computed and subsequently averaged across all trading days. Estimation uncertainty is visualized using 95% confidence intervals around the average realized variance.

1.4 Compute Realized Variance for All Sampling Frequencies

1.5 Summary Statistics

	delta	mean_rv	sd_rv	n_days	se_rv	ci_low	ci_high
0	1	0.000298	0.000626	4444	0.000009	0.000279	0.000316
1	2	0.000264	0.000563	4444	0.000008	0.000247	0.000280
2	5	0.000230	0.000472	4444	0.000007	0.000217	0.000244
3	10	0.000209	0.000448	4444	0.000007	0.000196	0.000222
4	15	0.000193	0.000412	4444	0.000006	0.000181	0.000205
5	30	0.000167	0.000350	4444	0.000005	0.000156	0.000177
6	60	0.000167	0.000385	4444	0.000006	0.000155	0.000178

1.6 Volatility Signature Plot



1.7 Interpretation

The volatility signature plot displays the average daily realized variance for AAPL at different sampling frequencies. Missing minute-level prices are handled by previous-tick interpolation, meaning that a missing price is replaced by the most recently observed transaction price. This implies a zero return during minutes with no new observed trade.

The error bars represent 95% confidence intervals and visualize the estimation uncertainty of the realized variance estimator.

At very high sampling frequencies, realized variance estimates may be inflated due to market microstructure noise, including bid-ask bounce, price discreteness, and other trading frictions. As the sampling interval increases, the influence of microstructure noise declines and the realized variance estimates typically stabilize.

The sampling frequency used in the remainder of the analysis will be selected based on the point where the volatility signature plot becomes relatively stable while still retaining a substantial amount of high-frequency information.